

Jesse Quiroga

Senior/Staff Software Engineer | Full-Stack & Cloud-Native

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SUMMARY

Senior/Staff Soft Engineer with **12+** years architecting scalable platforms at **Google** and **Slack**, specializing in machine learning systems, real-time messaging infrastructure, and advertising technology. Proven expertise building **end-to-end** solutions from React/Angular frontends to distributed backend services, with deep experience in AI/ML recommendation engines, real-time data processing, and **cloud-native architectures**. Led engineering teams to deliver high-impact products serving billions of users across **consumer** and **enterprise platforms**.

PROFESSIONAL EXPERIENCE



Staff Software Engineer

Remote

Apr 2019 - Present

- Led core features development for **Google's Smart Bidding platform** and **Performance Max campaign automation**, processing billions of real-time auction requests and implementing cross-channel advertising optimization across Search, YouTube, Display, and Shopping networks.
- Built distributed backend infrastructure using **Java Spring Boot** microservices, **Apache Kafka** message queues, and **PostgreSQL** databases, creating scalable auction processing systems that handle peak traffic during major shopping events and architected fraud-detection pipelines similar to financial risk engines.
- Developed microservices architecture with **Node.js** APIs, **C# .NET** Core services, **MongoDB** document storage, and **RabbitMQ** message brokers, implementing event-driven campaign automation with **GraphQL** query optimization for real-time performance tracking.
- Implemented AI/ML optimization engines using **Python**, **TensorFlow**, and reinforcement learning algorithms, developing neural networks for bid optimization, multi-armed bandit algorithms for campaign automation, generative AI for automated asset creation, and designed **HIPAA-compliant** ML workflows for sensitive data.
- Deployed cloud-native architecture on **Google Cloud Platform** with **Kubernetes** orchestration, **Docker** containers, **Cloud Functions**, **BigQuery** analytics, and **Terraform** infrastructure-as-code, building serverless systems with auto-scaling capabilities.
- Established comprehensive testing frameworks using **JUnit**, **PyTest**, **Jest**, **Cypress**, and **Playwright**, implementing unit testing, integration testing, A/B testing platforms, and automated ML model validation with continuous integration pipelines.
- Designed full-stack user interfaces with **React/TypeScript**, **Angular/RxJS**, **Material-UI**, and **D3.js** data visualization, creating intuitive campaign management dashboards, performance analytics tools, and enterprise advertiser workflow builders.
- Mentored **cross-functional engineering teams** across ML engineering, full-stack development, and DevOps, establishing coding standards for ML workflows and leading technical design reviews for high-impact advertising platform features.



Senior Software Engineer

Remote

Mar 2015 - Mar 2019

- Rebuilt **YouTube's recommendation engine** from collaborative filtering to deep neural networks using **Python** and **TensorFlow**, implementing two-stage pipeline architecture for candidate generation and ranking that serves personalized video recommendations to billions of users worldwide.
- Built scalable backend infrastructure with **Go** microservices on **Kubernetes**, **C++** serving systems, and **Java** experimentation frameworks, creating distributed architecture that processes petabytes of user interaction data while maintaining system reliability during viral traffic events.

- Developed **AI/ML recommendation pipeline** using **Python** Multi-gate Mixture-of-Experts architecture, **TensorFlow** neural networks, and content safety classifiers, implementing multitask learning systems that balanced engagement metrics with user satisfaction across billions of video recommendations.
- Deployed cloud-native platform on **Google Cloud Platform** with **Apache Beam**, **Cloud Dataflow**, **BigQuery** analytics, and auto-scaling **Kubernetes** clusters, building real-time feature engineering systems that generate video embeddings and process user behavior signals.
- Created recommendation interface systems with **JavaScript** frontend components, **React** dashboard tools, and **Redux** state management, developing A/B testing frameworks and analytics dashboards that enabled data-driven product decisions across concurrent recommendation experiments.
- Mentored **engineering teams** on machine learning systems architecture, established coding standards for recommendation algorithms, and led technical design reviews for content discovery and personalization platform features.

Slack Technologies

San Francisco, CA

Senior Software Engineer

Jun 2013 - Feb 2015

- Co-architected **Slack's messaging platform** during public beta launch using **WebSocket** technology and real-time communication protocols, building scalable chat infrastructure that scaled from thousands to millions of daily active users within 24 months.
- Built core messaging backend with **Node.js** APIs, **PostgreSQL** databases, and **Redis** caching layers, implementing persistent chat channels, direct messaging, and message delivery systems that handle billions of messages with sub-50ms latency.
- Developed integration platform backend using **PHP** services, **MySQL** storage, and **OAuth 2.0** authentication protocols, creating webhook systems and API gateways that enabled seamless connectivity with 150+ external services including Google Calendar and GitHub.
- Deployed scalable infrastructure on **AWS** with **Docker** containers, **Elastic Load Balancers**, and **RDS** databases, building auto-scaling systems that maintained 99.99% uptime across team workspaces during rapid user growth phases
- Implemented comprehensive testing frameworks using **Jest**, **Mocha**, and integration testing suites, establishing automated testing pipelines for message delivery, authentication systems, and third-party service integrations
- Mentored **engineering teams** on real-time messaging architecture, established coding standards for WebSocket implementations, and led technical design reviews for core platform scalability and security features.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, Golang, Swift, Kotlin, SQL, C/C++

Backend Development: Spring Boot, Node.js/Express, Python/Django/FastAPI, Golang, REST APIs/GraphQL, gRPC

Frontend Development: React, React Native, Next.js, Redux, Angular, TypeScript, Tailwind CSS, Material-UI, D3.js

Machine Learning & AI: TensorFlow, PyTorch, Neural Networks, Deep Learning, Recommendation Systems, NLP,

Databases: PostgreSQL, MySQL, MongoDB, Redis, BigQuery, Cassandra, DynamoDB, Elasticsearch

Cloud & DevOps: Google Cloud Platform, AWS, Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, CI/CD

Message & Streaming: Apache Kafka, RabbitMQ, Pub/Sub, WebSocket, Apache Beam, Event-Driven Architecture

Testing Frameworks: Jest, JUnit, PyTest, Cypress, Playwright, Mocha, A/B Testing, Integration Testing

Architecture Patterns: Microservices, Event-Driven Design, Domain-Driven Design, Serverless, Distributed Systems

EDUCATION

The University of Texas at Austin

Bachelor of Science in Computer Science

2009 - 2013